

Skills Summary

Comparing Distributions and Compound Events

Use this reference while completing your worksheet or when studying.

Skill 1 — Comparing centre and spread

Centre — where the data sit. Mean = total $\div$ how many. Median = the middle value once the data are in order.

Spread — how stretched the data are. Range = largest – smallest.

To compare two sets, quote one of each. Equal centres do not make equal distributions: the smaller range is the steadier set. One unusually large value drags the mean but leaves the median alone, so report the median when a set has an outlier.

Example: Find the mean and the range of 2, 4, 6, 8, 10.

Step 1. A comparison needs one centre number and one spread number, so I compute the mean for centre and the range for spread.

Step 2. The five values total thirty, and five shares of that total give the mean; the largest minus the smallest gives the range.

Mean = 6, range = 8.

Try it: Find the mean and the range of 3, 4, 5, 6, 7.

check: mean 5, range 4

Skill 2 — Reading a data display

Method: read the scale before reading any bar. Then take the question one word at a time: *how many* for a single category, *altogether* for the total, *how many more* for a difference, *what fraction* for a category over the total.

Example: A tally of counters gives Red 7, Blue 4, Green 9. Find the total, and the number of blue counters.

Step 1. Two different questions on one display: a total needs every category, a single count needs only its own row.

Step 2. Adding the three categories gives the total, while the blue row is read straight off the display.

Total = 20, blue = 4.

Try it: A pet survey gives Cats 6, Dogs 11, Fish 3. Find the total, and the number of dogs.

check: total 20, dogs 11

Skill 3 — Compound-event probability

Rule: $P(\text{event}) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$, then reduce.

For two things happening together, the total is the first count times the second count — a table of all outcomes shows why. Count the favourable cells in that table. When something is replaced, the second stage has the same counts as the first.

Example: A coin is flipped and a number cube is rolled. Find the probability of tails and an even number.

Step 1. Two stages happening together, so the total outcomes multiply — two coin results times six cube results.

Step 2. Favourable cells are tails paired with each of the three even faces, and that count over the total reduces.

$$\Rightarrow P = \frac{3}{12} = \frac{1}{4}$$

Try it: Two number cubes are rolled. Find the probability that the total is five. There are 4 favourable pairs out of 36.

$\frac{6}{1}$:peep